STAR MOTION AND PATTERNS

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STARS MAY SEEM TO BE FIXED to the celestial sphere, but in fact their positions are changing, albeit very slowly. There are two parts to this motion: a tiny, yearly wobble of a star's position in the sky, called parallax shift; and a continuous directional motion, called proper motion. To record the motion of stars, and properties such as their color and brightness, each star needs a name. Naming systems and catalogs have their roots in the constellations, which were invented to describe the patterns formed by stars in the sky.

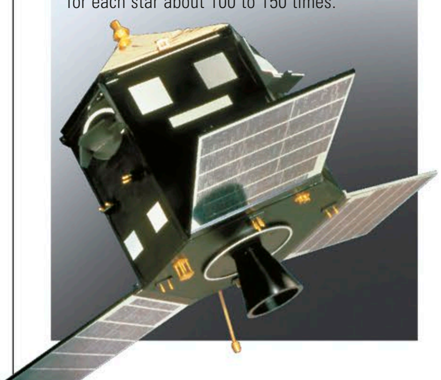
STAR COLORS
Although at first glance they all look white, stars differ in their colors—that is, in the mixture of light wavelengths they emit. This is a long-exposure photograph of the bright stars of Orion, taken while changing the camera's focus. Each star looks white when sharply focused, but when its light is spread out, its true color is revealed.

EXPLORING SPACE

HIPPARCOS

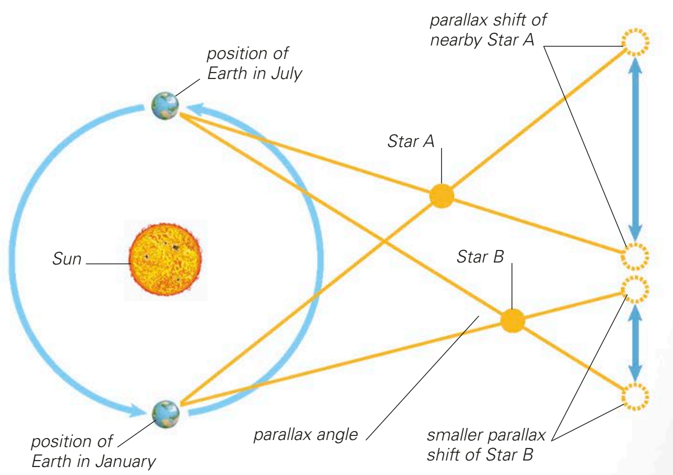
Hipparcos was a European Space Agency satellite that between 1989 and 1993 performed surveys of the stars. Its name is short for High Precision Parallax Collecting Satellite and was chosen to honor the Greek astronomer Hipparchus. Its mission has resulted in two catalogs. The Hipparcos catalog records the position, parallax, proper motions, brightness, and color of over 118,000 stars to a high level of precision. The Tycho catalog records over 1 million stars with measurements of lower accuracy.

HIPPARCOS SATELLITE
The satellite spun slowly in space, scanning strips of the sky as it rotated. It recorded data for each star about 100 to 150 times.



PARALLAX SHIFT

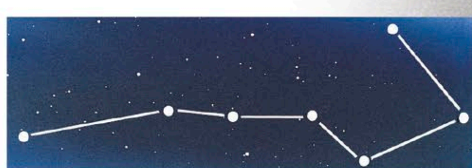
Parallax shift is an apparent change in the position of a relatively close object against a more distant background as the observer's location changes. When an observer takes two photographs of a nearby star from opposite sides of Earth's orbit around the Sun, the star's position against the background of stars moves slightly. When the observer measures the size of this shift, knowing the diameter of Earth's orbit, she or he can calculate the star's distance using trigonometry. Until recently, this technique was limited to stars within a few hundred light-years of Earth, because the shifts of distant stars were too small to measure accurately. However, by using accurate instruments carried in satellites, much greater precision is possible: those carried in the Hipparcos satellite (see panel, left) have allowed calculation of star distances up to a few thousand light-years from Earth. More recently, a satellite named Gaia has been measuring the distances of stars as far away as the center of our galaxy and beyond.



MEASURING DISTANCE USING PARALLAX
When Star A is observed from opposite sides of Earth's orbit, its apparent shift in position is greater than that of more distant Star B. From the shift, an observer can calculate the parallax angle between the star and the two positions of Earth. The star's distance can be determined from this angle.

PROPER MOTION OF STARS

All stars in our galaxy are moving at different velocities relative to the solar system, to the galactic center, and to each other. This motion gives rise to an apparent angular movement across the celestial sphere called a star's proper motion—measured in degrees per year. Most stars are so distant that their proper motions are negligible. About 200 have proper motions of more than 1 arc-second a year—or 1 degree of angular movement in 3,600 years. Barnard's star (see p.381) has the fastest proper motion, moving at 10.3 arc-seconds per year. It takes 180 years to travel the diameter of a full moon in the sky. If astronomers know both the proper motion of a star and its distance, they can calculate its transverse velocity relative to Earth—that is, its velocity at right angles to the line of sight from Earth. The other component of a star's velocity relative to Earth is called its radial velocity (its velocity toward or away from Earth), measured by shifts in the star's spectrum (see p.35).



THE BIG DIPPER IN 100,000 BCE

THE BIG DIPPER IN 2000 CE

THE BIG DIPPER IN 100,000 CE

CHANGING SHAPE
The shape of the star pattern known as the Big Dipper gradually changes due to the proper motions of its stars. Five stars are moving in unison as a group, but the two stars on the ends are moving independently.

INTRODUCTION